

3(a)	$p = 1000 \times 9.81 \times 7.0 \times 10^{-2}$ or $1000 \times 9.81 \times 1.9 \times 10^{-2}$	C1
	$\Delta p = 1000 \times 9.81 \times (7.0 \times 10^{-2} - 1.9 \times 10^{-2})$ or $686 - 186$ $= 500 \text{ Pa}$	A1
3(b)	$F = pA$ or $(\Delta)F = \Delta p \times A$	C1
	upthrust $= 500 \times (5.1 \times 10^{-2})^2 = 1.3 \text{ N}$ or upthrust $= (686 - 186) \times (5.1 \times 10^{-2})^2 = 1.3 \text{ N}$ or upthrust $= 1000 \times 9.81 \times 5.1 \times 10^{-2} \times (5.1 \times 10^{-2})^2 = 1.3 \text{ N}$	A1
3(c)	force $= 4.0 - 1.3$ $= 2.7 \text{ N}$	A1

Question	Answer	Marks
3(d)	extension/x/e = $2.7/30$	C1
	= 0.09 (m) or 9 (cm)	C1
	height above surface = $9 - 7$ = 2 cm	A1
3(e)(i)	mass = $4.0/9.81$	C1
	acceleration = $2.7/(4.0/9.81)$ = 6.6 ms^{-2}	A1
3(e)(ii)	viscous force <u>increases</u> (and then becomes constant)	M1
	(weight and upthrust constant so) acceleration decreases (to zero)	A1